

AURORA SENTINEL FINAL PROJECT REVIEW REPORT SUMMARY

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Overview

This report summarizes information extracted from the authorized selected document.

Key Findings

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1. [Page 1 | native] AURORA-SENTINEL - Final Project Review Page 1 AURORA-SENTINEL Final Project Architecture, Security, Reliability and Code Review Repository reviewed:
KrishnaRajput07/AURORA-SENTINEL Default branch: main Repository state reviewed: September 12, 2026 Overall verdict: Advanced hackathon / prototype system with strong feature breadth, but not production-ready.
 2. The largest issues are architecture drift, inconsistent risk-score semantics, missing backend security controls, and deployment inconsistencies.
 3. Prepared from repository structure, source files, configuration, tests, deployment files, and the project's own design/requirements documents.
 4. Binary model files are treated as artifacts; the report does not claim semantic inspection of model weights.
 5. [Page 2 | native] AURORA-SENTINEL - Final Project Review Page 2 1.
 6. Executive Summary AURORA-SENTINEL is an AI-assisted surveillance platform designed to ingest live or uploaded video, detect potentially dangerous activity, score risk, use vision-language models for contextual verification, generate operator alerts, capture relevant video, retain/archive evidence, and expose search and chatbot interfaces to operators.
 7. The repository is substantial and modular.
 8. It contains a FastAPI backend, a React dashboard, YOLO-based detection and pose pipelines, fire/smoke and weapon detectors, temporal risk scoring, VLM integrations, semantic search, video retention, alert lifecycle management, a chatbot/agent, Docker deployment material, and multiple Kiro specifications.
 9. The principal problem is not lack of functionality.
 10. It is architectural consolidation.
 11. Multiple versions of the scoring and AI orchestration design coexist, and some current runtime paths do not implement the same policy described by the current specifications.
 12. This creates correctness, operational, and maintainability risk.
 13. Area Assessment Product concept Strong Feature breadth Strong Backend structure Good ML engineering Good prototype AI/VLM integration Interesting but fragmented Risk scoring Needs redesign/consolidation Authentication / authorization Critical gap Deployment consistency Poor Testing Moderate Production readiness No 2.
 14. Project Architecture and Components The system can be summarized as follows: Camera / Uploaded Video -> Frame ingestion / scheduling -> YOLO object detection + pose estimation + specialist detectors -> Risk engine and temporal validation -> VLM contextual analysis (Ollama / Gemini / optional Qwen / optional Nemotron) -> Alert + evidence capture -> Postgres/SQLite + filesystem storage + semantic search -> React operator dashboard + chatbot / agent Major subsystems Backend/API: FastAPI routers for alerts, analytics, video, archive, intelligence, settings, smart-bin, chatbot and live/VLM WebSockets.
 15. Detection: Unified YOLO detector with general objects, human pose, fire/smoke, vehicle and weapon detection, plus a simple tracker.
 16. Risk engine: Multi-factor scoring using aggression, weapons, fire/smoke, proximity, loitering, unattended objects, crowd density, context, temporal history and escalation rules.
 17. VLM/AI: Ollama primary path, Gemini fallback, separate Qwen2-VL AI layer, optional Nemotron

verification, and a chatbot/agent path.

18. [Page 3 | native] AURORA-SENTINEL - Final Project Review Page 3 Evidence: Rolling/chunked video capture, clip extraction/transcoding, Smart Bin retention, database clip records.

19. Search: ChromaDB + SentenceTransformer vector search with metadata/keyword fallbacks and timeline-aware ranking.

20. Frontend: React dashboard with live feed, alert queue, analytics, archive, intelligence panel, Smart Bin, settings and chatbot UI.

21. 3.

22. Strengths and Positive Engineering Decisions · Good service decomposition.

23. ML, risk scoring, VLM providers, storage, retention, search and chatbot concerns are mostly separated into dedicated modules. · The detector goes beyond object detection and includes pose estimation, fire/smoke, vehicles, weapons, blur detection and tracking. · The risk engine includes temporal validation, keypoint velocity history, proximity, grappling and calibration rather than relying only on a single frame. · The AI layer includes model availability tracking and a circuit-breaker style cooldown after repeated provider failures. · Search is designed for forensic use, including vector retrieval, metadata filtering, timeline search and cross-video search. · Video retention is treated as a first-class concern rather than an afterthought. · Automated tests exist across several backend features and at least one frontend component. · The project explicitly documents requirements and acceptance criteria.

24. This is useful for future engineering consolidation.

25. [Page 4 | native] AURORA-SENTINEL - Final Project Review Page 4 4.

26. Critical Findings C1 - Inconsistent final-score policy The repository contains incompatible definitions of the final score.

27. One implementation uses a weighted score (30% ML / 70% AI), while the enhanced fight-detection requirements define Final_Score as MAX(ML_Score, AI_Score), and another architecture description specifies a hybrid.

28. These policies can produce radically different alerts.

29. For example, ML=90 and AI=20 gives 41 under the weighted formula but 90 under MAX.

30. Recommended action: Fix: choose one canonical policy; for operator safety, MAX(ML, AI) is the clearest conservative policy.

31. Persist both component scores.

32. C2 - Offline video path does not actually use the intended two-tier scoring service The offline video processor contains the TwoTierScoring integration as commented-out code and directly calls RiskScoringEngine.calculate_risk().

Sources

1. AURORA_SENTINEL_Final_Project_Review_Report.pdf